

Improving asthma care through patient-reported outcome measure optimization



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Background

- Optimizing asthma control is paramount in ensuring patients with asthma live a normal life but is often overestimated in clinical practice.¹
- To assess asthma control, guidelines recommend evaluating current asthma symptoms and the risk of future exacerbations.²
- The Asthma Impairment and Risk Questionnaire (AIRQ) is a validated patient-reported outcome measure (PROM) of asthma control that encompasses both domains.³
- Health-system specialty pharmacies (HSSPs) often rely on PROMs, such as the AIRQ, to monitor the effectiveness and safety of medications. Selecting PROMs that optimize patient outcomes while maintaining operational efficiency and data quality is essential in these settings.

Objective

To describe the integration of a new PROM aimed at optimizing care for patients with asthma into an existing HSSP clinical management process.

Methods

Study Design

- This descriptive study details the process of selecting, implementing, and evaluating a new PROM for the management of patients with asthma in the HSSP setting.

PROM Selection Process

- In April 2024, CPS Solutions, LLC (CPS) clinical subcommittees evaluated available asthma-specific PROMs and elected to transition to the AIRQ due to its ease of administration and ability to assess both exacerbation risk and symptom control.

HSSP Clinical Management Protocol

- All patients receiving an asthma medication from a CPS-managed HSSP complete the AIRQ via phone, text, or in-person after initial enrollment into HSSP services and every three to six months thereafter, depending on patient-specific parameters.

Data Analysis

- Descriptive statistics were used to report evaluation measures, and bivariate tests were used to compare baseline and follow-up measures.

Results

- In total, baseline and follow-up AIRQ scores from 911 patients across 20 health systems were included.
- The median time interval between baseline and follow-up scores was 5.0 months (IQR 3.5-6.1).

Figure 1. AIRQ Scores at Baseline and Follow-Up (N=911)

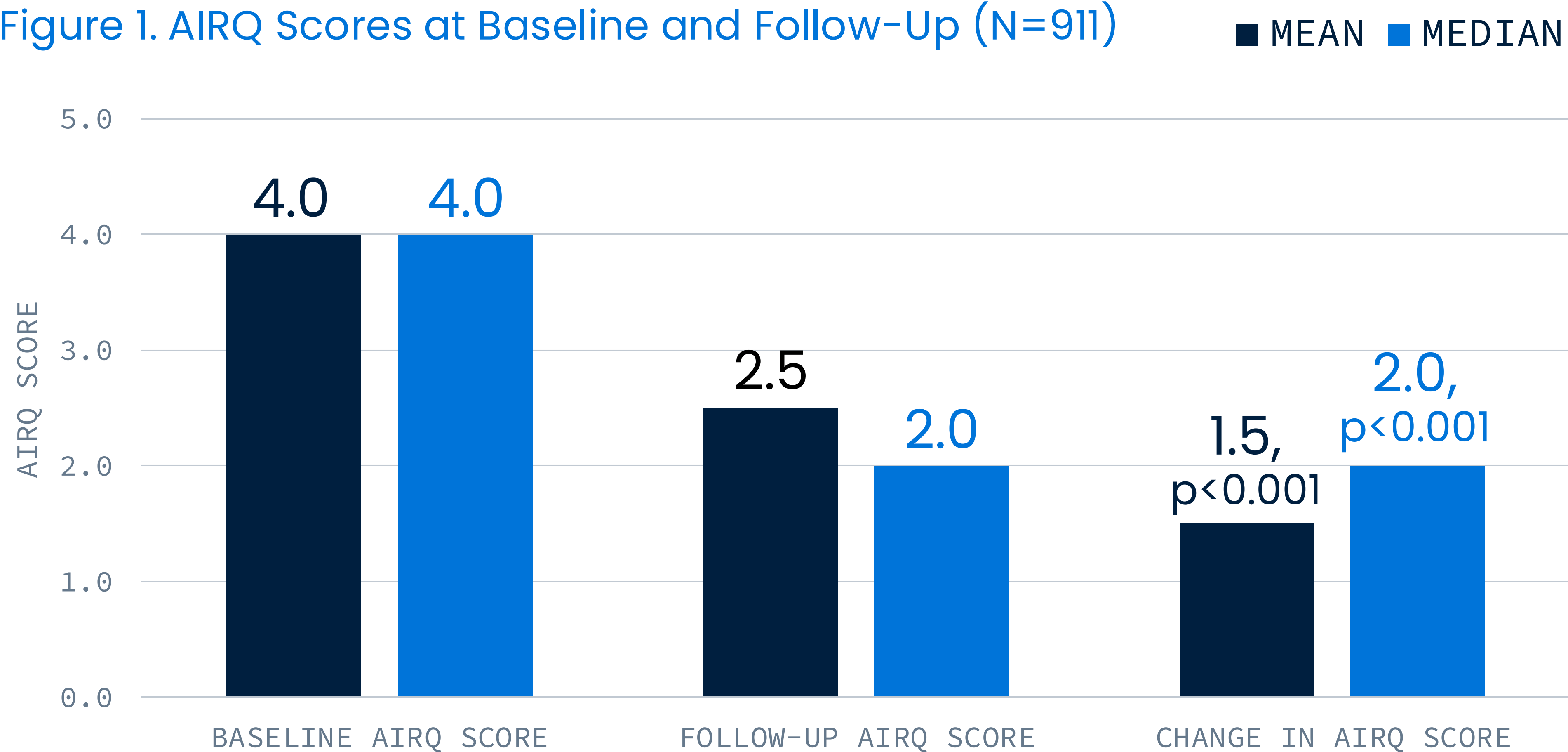
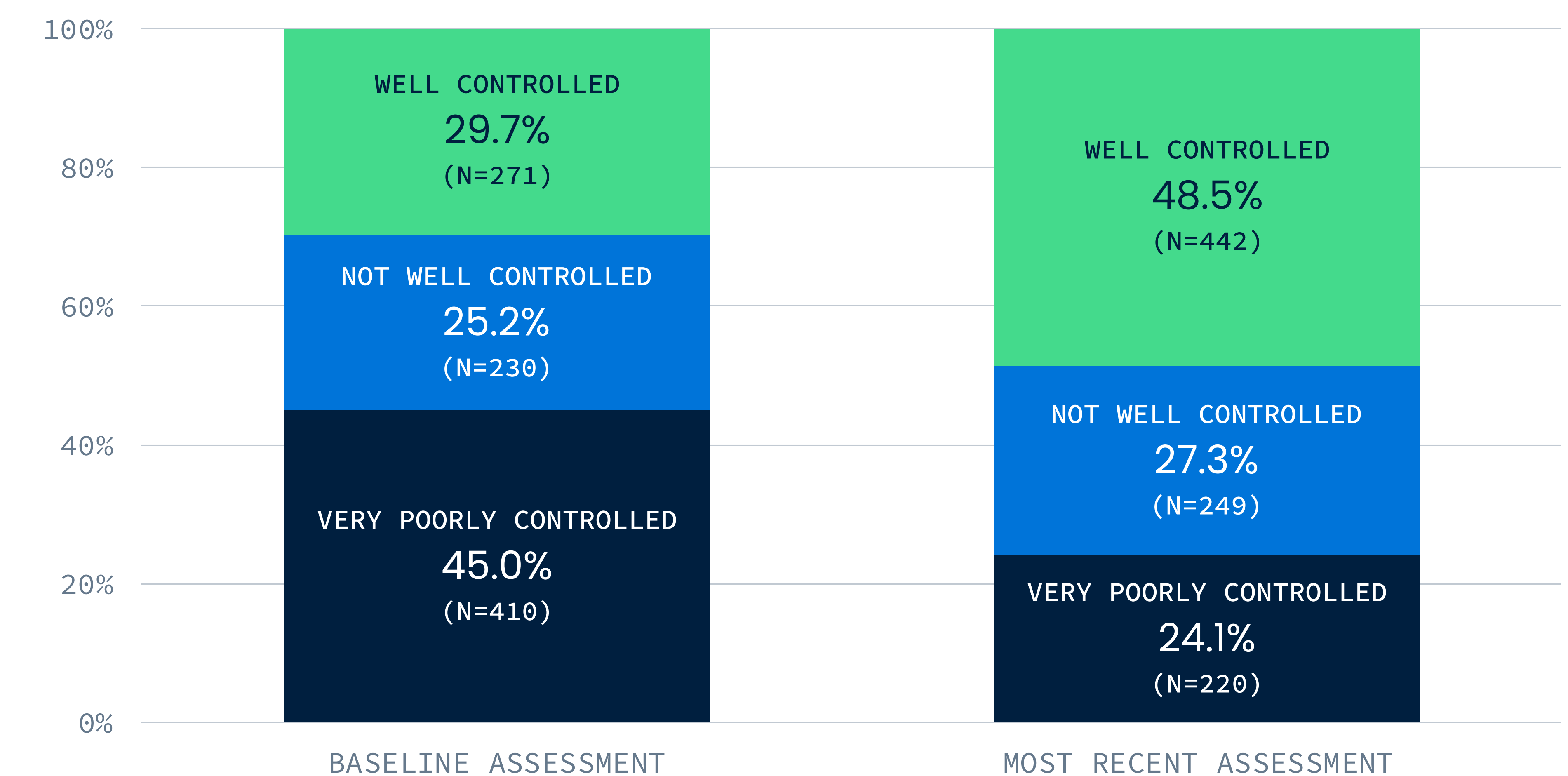


Figure 2: Change in Asthma Control at Baseline and Follow-Up (N=911)



Results Cont.

60%

Among patients with not well-controlled asthma or very poorly controlled asthma at baseline (n=640), 60% (n=387) demonstrated a clinically meaningful improvement in their AIRQ scores at follow-up, defined as a reduction of two or more points from baseline.

19%

The proportion of patients with well-controlled asthma increased by nearly 19%—from 29.7% at baseline to 48.5% at follow-up (p<0.001)—while the percentage of patients with very poorly controlled asthma decreased by approximately 21%, from 45.0% to 24.1% (p<0.001), over the same time period.

Discussion & Conclusion

- For optimized patient management, HSSPs should periodically evaluate utilized PROMs and transition to alternatives when needed.
- The AIRQ was successfully integrated into an existing HSSP clinical management process without significant disruption, demonstrating its feasibility in real-life practice.
- Utilization of the AIRQ has enabled a comprehensive assessment of asthma control by incorporating both current asthma symptoms and the risk of future exacerbations. This provides pharmacists with actionable data to support decision-making, including therapy optimization.

References

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